Module : 2

Previous Year Questions

Part - A

1) Derive expression for static pressure ratio and static temperature ratio across a normal shock wave

2) Derive Prandtl's relation for a flow through oblique shock.

3) Explain impossibility of a shock wave in subsonic flow using plot showing change of entropy across a normal shock wave versus upstream Mach number

4) Explain attached and detached shocks

5) How do you differentiate between weak and strong oblique shocks? Explain.

Part : B

1) Derive Prandtl's Meyer relation for a flow through normal shock.

2) Explain $\theta - \beta - M$ diagram

3) Derive Rankine - Hugoniot relation for normal shock

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* $$\frac{T_0}{T} = \left[1 + \left(\frac{\gamma-1}{2}\right) M^2\right]$$

* $$\frac{P_0}{P} = \left[1 + \left(\frac{\gamma-1}{2}\right) M^2\right]^{(\gamma/\gamma-1)}$$

* $$\frac{\rho_0}{\rho} = \left[1 + \left(\frac{\gamma-1}{2}\right) M^2\right]^{1/\gamma-1}$$

* Mach Number downstream of normal shock

$$M_y = \left[\frac{\frac{2}{\gamma-1} + M_x^2}{\frac{2\gamma}{\gamma-1} M_x^2 - 1} \right]^{1/2} \longrightarrow ①$$

[From data book]

* Static Pressure Ratio

$$\boxed{\rho c^2 = \gamma P M^2}$$

$F_x = F_y$

ie; $\rho c^2 = \frac{P}{RT} c^2$

$$P_x + \rho_x c_x^2 = P_y + \rho_y c_y^2$$

$= \frac{\gamma P c^2}{\gamma R T} = \underline{\underline{\gamma P M^2}}$

$$\therefore P_x + \gamma P_x M_x^2 = P_y + \gamma P_y M_y^2$$

$$\boxed{\frac{P_y}{P_x} = \frac{1 + \gamma M_x^2}{1 + \gamma M_y^2}} \longrightarrow ②$$

(databook pg 5)

If we substitute equ① in equ②

$$\boxed{\frac{P_y}{P_x} = \frac{2\gamma}{\gamma+1} M_x^2 - \frac{\gamma-1}{\gamma+1}}$$

(databook pg: 5)

* Temperature Ratio

We know;

$$\frac{T_{ox}}{T_x} = 1 + \frac{\gamma-1}{2} M_x^2$$

$$\frac{T_{oy}}{T_y} = 1 + \frac{\gamma-1}{2} M_y^2$$

From adaibatic equ:

$$T_{0x} = T_{0y} = T_0$$

$$\therefore \quad \frac{T_y}{T_x} = \frac{T_{0x}}{T_x} \cdot \frac{T_y}{T_{0y}}$$

ie;
$$\boxed{\frac{T_y}{T_x} = \frac{1 + \left(\frac{\gamma - 1}{2}\right) M_x^2}{1 + \left(\frac{\gamma - 1}{2}\right) M_y^2}} \longrightarrow (3)$$

(Databook Page 5)

if we substitute equ(1) in equ(3), we get:

$$\boxed{\frac{T_y}{T_x} = \frac{\left(1 + \frac{\gamma - 1}{2} M_x^2\right)\left(\frac{2\gamma}{\gamma - 1} M_x^2 - 1\right)}{\frac{(\gamma + 1)^2}{2(\gamma - 1)} M_x^2}}$$

(Databook Page 5)

* Density Ratio

$$P_x = \rho_x R T_x$$

$$P_y = \rho_y R T_y$$

$$\frac{P_y}{P_x} = \frac{\rho_y}{\rho_x} \frac{T_y}{T_x}$$

$$\therefore \quad \boxed{\frac{\rho_y}{\rho_x} = \frac{P_y}{P_x} \cdot \frac{T_x}{T_y}}$$

(databook pg:5)

* Velocity ratio

$$\rho_x c_x = \rho_y c_y$$

$$\boxed{\frac{c_x}{c_y} = \frac{\rho_y}{\rho_x}}$$

* Stagnation Pressure Ratio

$$\boxed{\frac{P_{oy}}{P_{ox}} = \frac{P_{oy}}{P_y} \cdot \frac{P_y}{P_x} \cdot \frac{P_x}{P_{ox}}}$$ (databook)

* Change in entropy

$$\boxed{\frac{\Delta s}{R} = \ln \frac{P_{ox}}{P_{oy}}}$$ (databook)

Basics

* $\frac{c_p}{c_v} = \gamma$
* $R_u = 8314 \ J/kg\,K$
* $a = \sqrt{\gamma R T}$
* $M = \frac{c}{a}$
* $R_{air} = 287 \ J/kg\,K$
* $\gamma_{air} = 1.4$
* $C_p = \frac{\gamma R}{\gamma - 1}$
* $C_v = \frac{R}{\gamma - 1}$

Problems

Q:1) The state of a gas ($\gamma = 1.3$, $R = 0.469$ kJ/kgK) upstream of a normal shock:

$$M_x = 2.5, \quad P_x = 2 \text{ bar}, \quad T_x = 275 \text{ K}$$

Calculate M_y, P_y, T_y and c_y of gas.

Ans:

* $$M_y = \left(\frac{\frac{2}{\gamma-1} + M_x^2}{\frac{2\gamma}{\gamma-1} M_x^2 - 1} \right)^{1/2} = \left[\frac{\frac{2}{1.3-1} + (2.5)^2}{\frac{2 \times 1.3}{1.3-1} (2.5)^2 - 1} \right] = \underline{\underline{0.492}}$$

* $$\frac{P_y}{P_x} = \frac{1 + \gamma M_x^2}{1 + \gamma M_y^2}$$

* $$P_y = 2 \times \left[\frac{1 + (1.3 \times 2.5^2)}{1 + (1.3 \times 0.492^2)} \right] = \underline{\underline{13.87 \text{ bar}}}$$

$$\frac{T_y}{T_x} = \frac{1 + \frac{\gamma-1}{2} M_x^2}{1 + \frac{\gamma-1}{2} M_y^2} = \frac{1 + \left(\frac{1.3-1}{2}\right) 2.5^2}{1 + \left(\frac{1.3-1}{2}\right) 0.49^2}$$

* $\therefore$ $T_y = \underline{\underline{513.9 \text{ K}}}$

* $$c_y = M_y \cdot \sqrt{\gamma R T_y}$$

$$= 0.492 \times \sqrt{1.3 \times 469 \times 513.9}$$

$$= \underline{\underline{275.16 \text{ m/s}}}$$

Q:2) A gas ($\gamma = 1.4$, $R = 0.287\ kJ/kgK$) at $M_x = 1.8$, $P_x = 0.8$ bar and $T_x = 373k$ passes through a normal shock. Determine density after shock.

Ans:

$$P_x = \rho_x R T_x$$

$$\rho_x = \frac{P_x}{R T_x} = \frac{0.8}{287 \times 373} = 0.747\ kg/m^3$$

From normal shock table, (ie; $\gamma = 1.4$ and $M = 1.8$)

At $M_x = 1.8$,

$$M_y = 0.616$$

$$\frac{P_y}{P_x} = 3.613$$

$$P_y = 3.613 \times 0.8 = \underline{\underline{2.8904}}\ \text{bar}$$

$$\frac{T_y}{T_x} = 1.532$$

$$T_y = 1.532 \times 373 = \underline{\underline{571.436\ K}}$$

$$\rho_y = \frac{P_y}{R T_y} = \frac{2.8904}{287 \times 571.436} = \underline{\underline{1.762\ kg/m^3}}$$

Now, for isentropic flow,

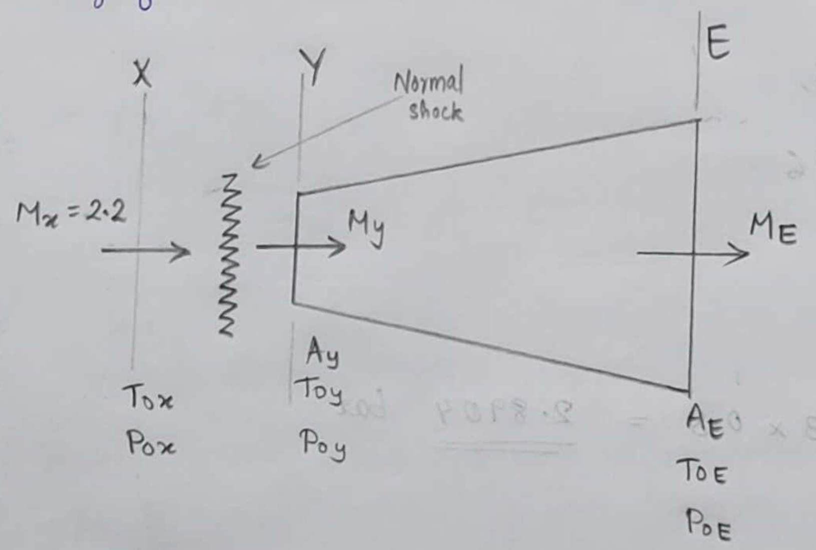
$$\frac{\rho_y}{\rho_x} = \left(\frac{P_y}{P_x}\right)^{1/\gamma}$$

ie; $$\rho_y = \rho_x \cdot \left(\frac{P_y}{P_x}\right)^{1/4} = 0.747 \times (3.613)^{1/1.4} = \underline{\underline{1.867\ kg/m^3}}$$

Q:3) The ratio of exit to entry area in a subsonic diffuser is 4. The Mach no. of a jet of air approaching the diffuser at $P_o = 1.013$ bar, $T = 290$ K is 2.2. There is a normal shock just outside diffuser entry. The flow in diffuser is isentropic. Determine at exit of diffuser:

(i) Mach no.
(ii) Temp
(iii) Pressure
(iv) What is stagnation pressure loss between initial and final stages of flow?

Ans:



$$\frac{A_E}{A_y} = 4$$

$M_x = 2.2$, $P_{ox} = 1.013$ bar, $T_x = 290$ K

From isentropic tables at $\gamma = 1.4$ & $M_x = 2.2$,

$$\frac{P_x}{P_{ox}} = 0.0935 \quad , \quad \frac{T_x}{T_{ox}} = 0.508$$

$$P_x = 0.0935 \times 1.013$$

$$P_x = 0.0947 \text{ bar}$$

$$T_{ox} = \frac{290}{0.508}$$

$$T_{ox} = 570.87 \text{ K}$$

From normal shock table,

At $M_x = 2.2$,

$M_y = 0.547$

$\frac{P_y}{P_x} = 5.480$, $\frac{T_y}{T_x} = 1.857$, $\frac{P_{oy}}{P_{ox}} = 0.628$

$P_y = 0.0947 \times 5.48$

$P_y = \underline{0.519 \text{ bar}}$

$T_y = 290 \times 1.857$

$T_y = \underline{538.53 \text{ K}}$

$P_{ox} = \frac{1.013}{0.628}$

$P_{ox} = \underline{1.613 \text{ bar}}$

From Isentropic table, at $M = 0.547$ and $\gamma = 1.4$,

$\frac{A_y}{A_y^*} = 1.259$; $\frac{P_y}{P_{oy}} = 0.82$

$P_{oy} = \frac{0.519}{0.82}$

$P_{oy} = \underline{0.636 \text{ bar}}$

We know,

$A_E^* = A_y^*$

$\frac{A_y}{A_E^*} = \frac{A_y}{A_y^*}$

Interpolation

$\frac{0.54 - 0.547}{0.547 - 0.55} = \frac{1.27 - x}{x - 1.255}$

$\frac{0.007}{0.003} = \frac{1.27 - x}{x - 1.255}$

$0.007x - 0.008785 = 0.00381 - 0.003x$

$0.01x = 0.012595$

$x = \underline{1.259}$

Now, $\frac{A_E}{A_E^*} = \frac{A_y}{A_E^*} \times \frac{A_E}{A_y}$

$= 1.259 \times 4$

$\frac{A_E}{A_E^*} = \underline{5.04}$

Now from isentropic tables, ~~find value of~~

$\frac{A_E}{A_E^*} = 5.04$ is at $M_E = \underline{0.116}$

$$\frac{P_E}{P_{0E}} = 0.992$$

We know $P_{0E} = P_{0y}$,

$$\therefore \; P_E = 0.992 \times P_{0y}$$

$$= 0.992 \times 0.636$$

$$P_E = \underline{\underline{0.629 \text{ bar}}}$$

NOTE

* For Isentropic flow,

$$T_{0x} = T_{0y}$$

$$P_{0x} = P_{0y}$$

* For normal shock,

$$T_{0x} = T_{0y}$$

$$P_{0x} \neq P_{0y}$$

Stagnation pressure loss $= P_{0x} - P_{0y}$

$$= 1.013 - 0.636$$

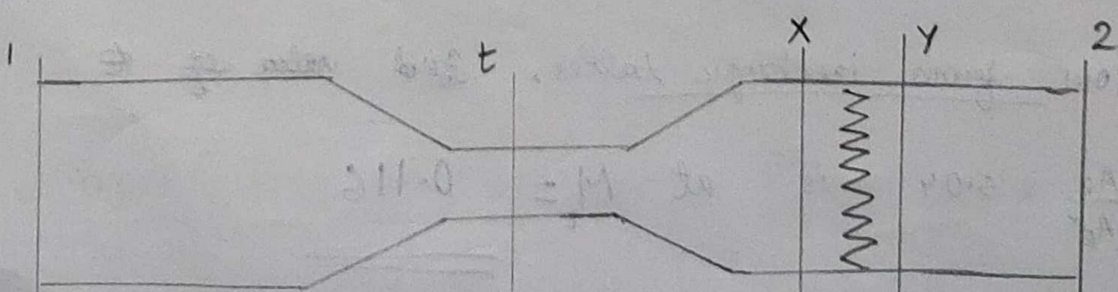
$$= \underline{\underline{0.377 \text{ bar}}}$$

Q:4) A normal shock occurs in the diverging section of a convergent-divergent air nozzle. The throat area is $\frac{1}{3} \times$ (exit area) and the static pressure at exit is 0.4 times the stagnation pressure at entry. The flow is throughout isentropic ~~except~~ except through shock. Determine:

a) Mach numbers M_x and M_y

b) Static pressure

c) Area of cross section of nozzle at the section of the nozzle where normal shock occurs.



$A_t = \frac{1}{3} A_2$

$P_2 = 0.4\, P_{01}$

$P_{01} = P_{0t} = P_{0x}$

$P_{0y} = P_{02}$

$A_1^* = A_t^* = A_t = A_x^*$

$A_y^* = A_2^*$

$\therefore \quad A_1^* P_{01} = A_2^* P_{02}$

$\frac{A_1^*}{A_2^*} = \frac{P_{02}}{P_{01}}$, also $\frac{A_t}{A_2^*} = \frac{P_{02}}{P_{01}}$

$\frac{\dot{m}\sqrt{T_0}}{A^* P_0} = \text{constant}$

For constant $\dot{m}$ & T_0,

$A^* P_0 = \text{constant}$

Now,

$\frac{A_2}{A_t} = 3$

$\frac{A_2 \times A_2^*}{A_2^* \times A_t} = 3$

$\therefore \quad \frac{A_2}{A_2^*} \times \frac{P_{01}}{P_{02}} = 3$

Sub: $P_2 = 0.4\, P_{01}$

$\cancel{\therefore \quad \frac{A_2}{A_2^*} \times \frac{P_{01}}{0.4\, P_{01}} = 3}$

$\cancel{\frac{A_2}{A_2^*} = 1.2}$

$\frac{A_2}{A_2^*} \times \frac{P_2}{0.4 \times P_{02}} = 3$

$\therefore \quad \frac{A_2}{A_2^*} \cdot \frac{P_2}{P_{02}} = 1.2$

From isentropic table,

search for $\frac{A_2}{A_2^*} \frac{P_2}{P_{02}} = 1.2$,

we get $M_2 = 0.47$

Also, $\frac{P_2}{P_{02}} = 0.859$

substitute; $P_2 = 0.4\, P_{01}$

$\therefore \frac{0.4\, P_{01}}{P_{02}} = 0.859$

$\frac{P_{02}}{P_{01}} = \underline{\underline{0.466}}$

$\therefore \frac{P_{0y}}{P_{0x}} = \underline{\underline{0.466}}$

From normal shock table, at $\frac{P_{0y}}{P_{0x}} = 0.466$

$M_x = 2.58$

$M_y = 0.506$

$\frac{P_{0y}}{P_x} = 9.145$

$P_x = \frac{P_{0y}}{9.145} = \frac{P_{02}}{9.145}$

$\therefore P_x = \frac{0.466\, P_{01}}{9.145}$

From isentropic table

$M_x = 2.58$

$A_x^* = A_t$

$\frac{A_x}{A_x^*} = 2.842$

$\therefore \; A_x = \underline{\underline{2.842 \, A_t}}$